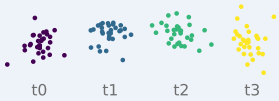
Optimal-transport conditional flow matching for unpaired single-cell snapshots

1 Unpaired snapshots



cells × genes at t_0 … t_m
per (condition, replicate)
no cell tracking, no PCA

2 Optimal transport

exact minibatch OT
(Hungarian assignment)
within one biological unit
chained $t_0 \to t_1 \to \ldots \to t_m$
= pseudo-trajectories

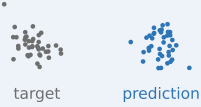
3 Spline paths

natural cubic spline per
pseudo-trajectory
in biological time
state x(u), velocity ẋ(u)

4 Autonomous field

v(x, c): MLP with SiLU
condition embedding, no time input
loss $\|v(x(u), c) - \dot{x}(u)\|^2$
Adam · EMA 0.999 · clip 1.0

5 Predict and evaluate



RK4 over $\Delta t = t_b - t_a$
20 steps per time unit
energy distance (U-statistic)
controls · permutation null
5 seeds, paired CI95

Design rules

Two clocks, never mixed:
the flow coordinate $s \in [0, 1]$
is a training device; biological
time (hours, days, stages) is data.
The field is autonomous and is
integrated over the biological
horizon; time will enter only
through regulatory gates (Phase 3).

Biological units are never split:
couplings live inside one
(condition, replicate) group; whole
time points and replicates are held out.